

Flaxseed supplementation improved insulin resistance in obese glucose intolerant people: a randomized crossover design

Background Obesity leads to an increase in inflammation and insulin resistance. This study determined antioxidant activity of flaxseed and its role in inflammation and insulin resistance in obese glucose intolerant people. **Methods** Using a randomized crossover design, nine obese glucose intolerant people consumed 40 g ground flaxseed or 40 g wheat bran daily for 12 weeks with a 4-week washout period. Plasma inflammation biomarkers (CRP, TNF- α , and IL-6), glucose, insulin, and thiobarbituric acid reactive substance (TBARS) were measured before and after of each supplementation. **Results** Flaxseed supplementation decreased TBARS ($p = 0.0215$) and HOMA-IR ($p = 0.0382$). Flaxseed or wheat bran supplementation did not change plasma inflammatory biomarkers. A positive relationship was found between TBARS and HOMA-IR ($r = 0.62$, $p = 0.0003$). **Conclusions** The results of the study weakly support that decreased insulin resistance might have been secondary to antioxidant activity of flaxseed. However, the mechanism(s) of decreased insulin resistance by flaxseed should be further determined using flaxseed lignan.